

SOC-5811 Week 10: Panel Data

Nick Graetz

University of Minnesota, Department of Sociology

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1/57



PANEL DATA

- ▶ What is “panel data?”



PANEL DATA

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- ▶ **Panel data:** A data set where we observe the same units (e.g., individuals, firms, countries, schools) over more than one time period.



PANEL DATA

- ▶ What is “panel data?”
- ▶ **Panel data:** A data set where we observe the same units (e.g., individuals, firms, countries, schools) over more than one time period.
- ▶ Walk through data examples: when is longitudinal data panel data? Repeated observations of the same variable.



PANEL DATA

Basic linear regression model:

$$y_i = \beta_0 + \beta_1 X_i$$

Linear regression model with panel data:

$$y_{i,t} = \beta_0 + \beta_1 X_{i,t}$$



PANEL DATA

- ▶ Let's walk through an example with data: traffic deaths and beer taxes by state/year.



PANEL DATA

```
dim(Fatalities)
```

```
## [1] 336 34
```

```
Fatalities %>% select(state) %>% table()
```

```
## state
## al az ar ca co ct de fl ga id il in ia ks ky la me md ma mi mn ms mo mt ne nv
## 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7
## nh nj nm ny nc nd oh ok or pa ri sc sd tn tx ut vt va wa wv wi wy
## 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7 7
```

```
Fatalities %>% select(year) %>% table()
```

```
## year
## 1982 1983 1984 1985 1986 1987 1988
## 48 48 48 48 48 48 48
```

PANEL DATA

- ▶ This is called “balanced” panel data. Why? What would be “unbalanced” panel data?



PANEL DATA

- ▶ This is called “balanced” panel data. Why? What would be “unbalanced” panel data?
- ▶ What shape is this dataset with respect to time?



PANEL DATA

```
head(Fatalities)
```

```
## # A tibble: 6 x 34
##   state year  spirits unemp income emmpop beertax baptist mormon drinkage dry
##   <fct> <fct>   <dbl> <dbl>   <dbl> <dbl>   <dbl>   <dbl>   <dbl>   <dbl> <dbl>
## 1 al     1982     1.37 14.4  10544.   50.7    1.54    30.4   0.328    19   25.0
## 2 al     1983     1.36 13.7  10733.   52.1    1.79    30.3   0.343    19   23.0
## 3 al     1984     1.32 11.1  11109.   54.2    1.71    30.3   0.359    19   24.0
## 4 al     1985     1.28  8.90  11333.   55.3    1.65    30.3   0.376   19.7   23.6
## 5 al     1986     1.23  9.80  11662.   56.5    1.61    30.3   0.393    21   23.5
## 6 al     1987     1.18  7.80  11944.   57.5    1.56    30.2   0.411    21   23.8
## # i 23 more variables: youngdrivers <dbl>, miles <dbl>, breath <fct>,
## #   jail <fct>, service <fct>, fatal <int>, nfatal <int>, sfatal <int>,
## #   fatal1517 <int>, nfatal1517 <int>, fatal1820 <int>, nfatal1820 <int>,
## #   fatal2124 <int>, nfatal2124 <int>, afatal <dbl>, pop <dbl>, pop1517 <dbl>,
## #   pop1820 <dbl>, pop2124 <dbl>, milestot <dbl>, unempus <dbl>,
## #   emmpopus <dbl>, gsp <dbl>
```

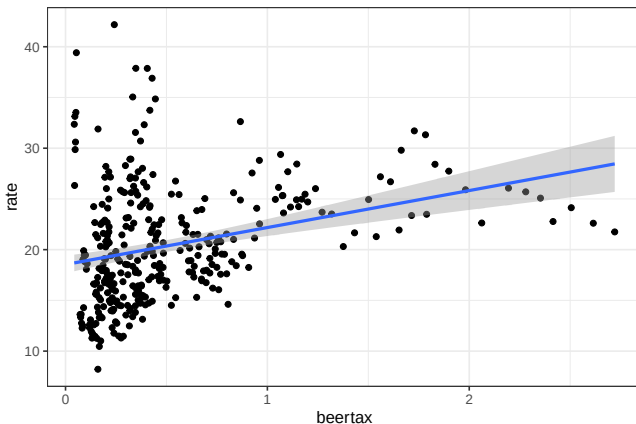
PANEL DATA

$$\text{Rate}_{i,t} = \beta_0 + \beta_1 \text{Tax}_{i,t}$$



PANEL DATA

All years (pooled model):



PANEL DATA

```
mod <- lm(rate ~ beertax, data=Fatalities)
summary(mod)
```

```
##
## Call:
## lm(formula = rate ~ beertax, data = Fatalities)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.9060  -3.7768  -0.9436   2.8548  22.7643
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  18.5331     0.4357  42.539 < 2e-16 ***
## beertax       3.6461     0.6217   5.865 1.08e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.437 on 334 degrees of freedom
## Multiple R-squared:  0.09336,    Adjusted R-squared:  0.09065
## F-statistic: 34.39 on 1 and 334 DF,  p-value: 1.082e-08
```

PANEL DATA

We could just think of panel data as several samples and several regression models:

$$\text{Rate}_{i,1988} = \beta_0 + \beta_1 \text{Tax}_{i,1988}$$

$$\text{Rate}_{i,1982} = \beta_0 + \beta_1 \text{Tax}_{i,1982}$$

PANEL DATA

$$\text{Rate}_{i,1988} = \beta_0 + \beta_1 \text{Tax}_{i,1988}$$

$$\text{Rate}_{i,1982} = \beta_0 + \beta_1 \text{Tax}_{i,1982}$$

$$\text{Rate}_{i,1988} - \text{Rate}_{i,1982} = \beta_0 + \beta_1 (\text{Tax}_{i,1988} - \text{Tax}_{i,1982})$$

We have now made this a “within” estimator: we are predicting within-unit changes.

PANEL DATA

```
Fatalities_wide <- Fatalities %>%  
  filter(year %in% c(1982,1988)) %>%  
  pivot_wider(id_cols = state,  
              values_from = c(rate,beertax),  
              names_from = year) %>%  
  mutate(rate_change = rate_1988 - rate_1982,  
         tax_change = beertax_1988 - beertax_1982)
```


PANEL DATA

```
head(Fatalities_wide)
```

```
## # A tibble: 6 x 7
##   state rate_1982 rate_1988 beertax_1982 beertax_1988 rate_change tax_change
##   <fct>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 al         21.3        24.9        1.54        1.50         3.66      -0.0379
## 2 az         25.0        27.1        0.215       0.346         2.07       0.132
## 3 ar         23.8        25.5        0.650       0.525         1.63      -0.126
## 4 ca         18.6        19.0        0.107       0.0866        0.417     -0.0208
## 5 co         21.7        15.1        0.215       0.173        -6.69     -0.0416
## 6 ct         16.5        15.0        0.224       0.217        -1.50     -0.00713
```

PANEL DATA

```
mod <- lm(rate_change ~ tax_change, data=Fatalities_wide)
summary(mod)
```

```
##
## Call:
## lm(formula = rate_change ~ tax_change, data = Fatalities_wide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.2715  -0.9619   0.9212   2.2290   6.7745
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.7204     0.6064  -1.188   0.2410
## tax_change   -10.4097     4.1723  -2.495   0.0162 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.94 on 46 degrees of freedom
## Multiple R-squared:  0.1192, Adjusted R-squared:  0.1
## F-statistic: 6.225 on 1 and 46 DF, p-value: 0.01625
```

PANEL DATA

- ▶ What does this remind you of that we have covered already?



PANEL DATA

- ▶ What does this remind you of that we have covered already?
- ▶ Simpson's Paradox! What was the idea of Simpson's Paradox related to omitted variables?



PANEL DATA

- ▶ What does this remind you of that we have covered already?
- ▶ Simpson's Paradox! What was the idea of Simpson's Paradox related to omitted variables?
- ▶ There was some variable we were missing. If we looked *within* penguin species, we found the opposite relationship!

PANEL DATA

- ▶ What does this remind you of that we have covered already?
- ▶ Simpson's Paradox! What was the idea of Simpson's Paradox related to omitted variables?
- ▶ There was some variable we were missing. If we looked *within* penguin species, we found the opposite relationship!
- ▶ **Intercept shifts.**

CAUSAL INFERENCE

- ▶ What is the difference between Simpson's Paradox examples and what we are doing here?



CAUSAL INFERENCE

- ▶ What is the difference between Simpson's Paradox examples and what we are doing here?
- ▶ There is no new *observed variable* that we are adding.



CAUSAL INFERENCE

- ▶ What is the difference between Simpson's Paradox examples and what we are doing here?
- ▶ There is no new *observed variable* that we are adding.
- ▶ So what have we done? We have accounted for *unobserved variables*.



CAUSAL INFERENCE

- ▶ What is the difference between Simpson's Paradox examples and what we are doing here?
- ▶ There is no new *observed variable* that we are adding.
- ▶ So what have we done? We have accounted for *unobserved variables*.
- ▶ But what specific kind of unobserved variables?
Time-invariant

CAUSAL INFERENCE

► Panel data and causal inference

$$\text{Rate}_{i,1988} = \beta_0 + \beta_1 \text{Tax}_{i,1988} + \beta_2 Z_i$$

$$\text{Rate}_{i,1982} = \beta_0 + \beta_1 \text{Tax}_{i,1982} + \beta_2 Z_i$$

$$\text{Rate}_{i,1988} - \text{Rate}_{i,1982} = \beta_0 + \beta_1 (\text{Tax}_{i,1988} - \text{Tax}_{i,1982})$$

CAUSAL INFERENCE

- ▶ When might we run into problems?



REVIEW

- ▶ Pick up here on Wednesday



PANEL DATA

Panel data: A data set where we observe the same units (e.g., individuals, firms, countries, schools) over more than one time period.

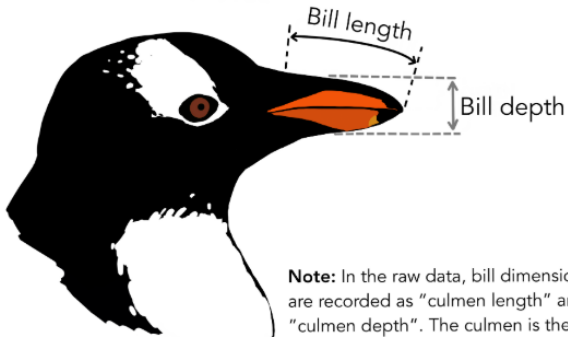
Linear regression model with cross-sectional data:

$$y_i = \beta_0 + \beta_1 X_i$$

Linear regression model with panel data:

$$y_{i,t} = \beta_0 + \beta_1 X_{i,t}$$

CROSS-SECTIONAL DATA



Note: In the raw data, bill dimensions are recorded as "culmen length" and "culmen depth". The culmen is the dorsal ridge atop the bill.

CROSS-SECTIONAL DATA

Cross-sectional data on penguin bill length and depth.

```
penguin_df <-  
  palmerpenguins::penguins %>%  
    na.omit()  
head(penguin_df)
```

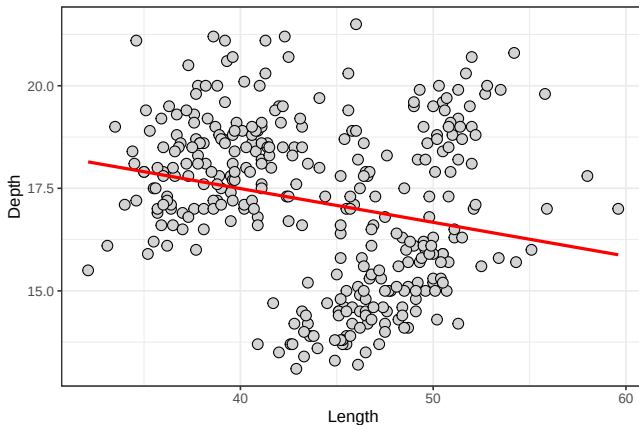
```
## # A tibble: 6 x 8  
##   species island   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g  
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int>  
## 1 Adelie  Torgersen         39.1          18.7          181          3750  
## 2 Adelie  Torgersen         39.5          17.4          186          3800  
## 3 Adelie  Torgersen         40.3          18           195          3250  
## 4 Adelie  Torgersen         36.7          19.3          193          3450  
## 5 Adelie  Torgersen         39.3          20.6          190          3650  
## 6 Adelie  Torgersen         38.9          17.8          181          3625  
## # i 2 more variables: sex <fct>, year <int>
```


CROSS-SECTIONAL DATA

$$\text{Depth}_i = \beta_0 + \boxed{\beta_1} \text{Length}_i$$

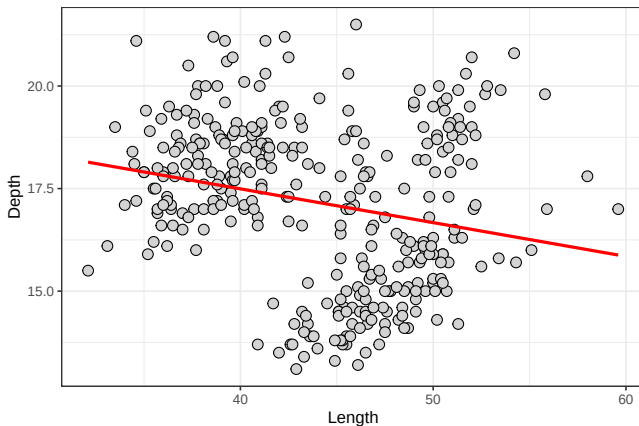
CROSS-SECTIONAL DATA

- What variation are we using to estimate the coefficient?



CROSS-SECTIONAL DATA

► Between-unit variation.



CROSS-SECTIONAL DATA

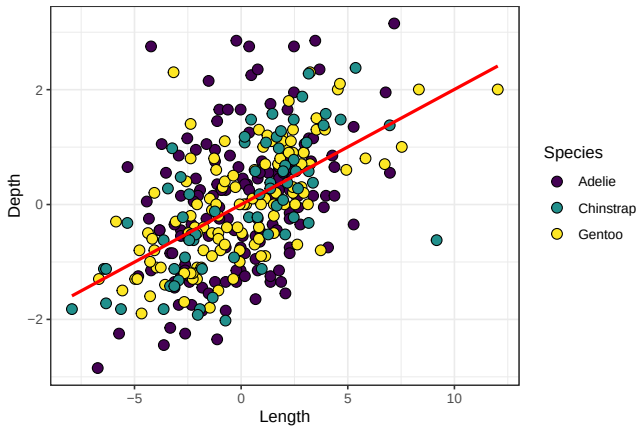
What variation are we using once we control for species?

$$\text{Depth}_i = \beta_0 + \boxed{\beta_1} \text{Length}_i + \beta_2 \text{Species}_i$$

CROSS-SECTIONAL DATA

```
penguin_df <- penguin_df %>%  
  group_by(species) %>%  
  mutate(bill_depth_mm_demeaned =  
    bill_depth_mm - mean(bill_depth_mm),  
    bill_length_mm_demeaned =  
    bill_length_mm - mean(bill_length_mm))
```

CROSS-SECTIONAL DATA



CROSS-SECTIONAL DATA

- What variation are we using to estimate this coefficient?



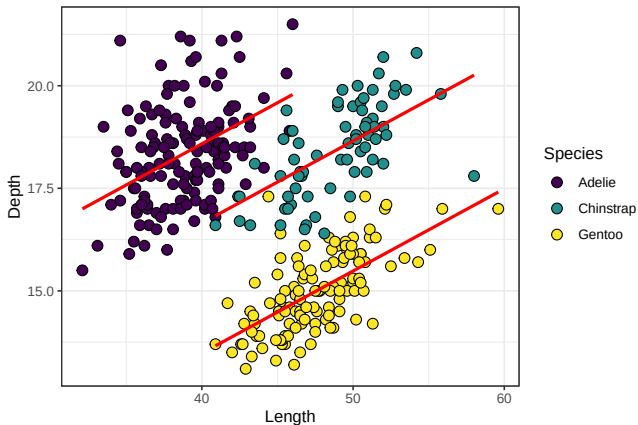
CROSS-SECTIONAL DATA

- ▶ What variation are we using to estimate this coefficient?
- ▶ Because we controlled for species, we are now using within-species variation. But with respect to unit, we are still using **between-unit** variation to estimate the coefficient.



CROSS-SECTIONAL DATA

- Intercept shift by species.



PANEL DATA

Panel data on traffic mortality rate and beer taxes.

```
head(Fatalities)
```

```
## # A tibble: 6 x 35
##   state year  spirits unemp income emppop beertax baptist mormon drinkage dry
##   <fct> <fct>   <dbl> <dbl>   <dbl> <dbl>   <dbl>   <dbl>   <dbl>   <dbl> <dbl>
## 1 al    1982     1.37 14.4  10544.  50.7    1.54    30.4  0.328    19  25.0
## 2 al    1983     1.36 13.7  10733.  52.1    1.79    30.3  0.343    19  23.0
## 3 al    1984     1.32 11.1  11109.  54.2    1.71    30.3  0.359    19  24.0
## 4 al    1985     1.28  8.90 11333.  55.3    1.65    30.3  0.376   19.7  23.6
## 5 al    1986     1.23  9.80 11662.  56.5    1.61    30.3  0.393    21  23.5
## 6 al    1987     1.18  7.80 11944.  57.5    1.56    30.2  0.411    21  23.8
## # i 24 more variables: youngdrivers <dbl>, miles <dbl>, breath <fct>,
## #   jail <fct>, service <fct>, fatal <int>, nfatal <int>, sfatal <fct>,
## #   fatal1517 <int>, nfatal1517 <int>, fatal1820 <int>, nfatal1820 <int>,
## #   fatal2124 <int>, nfatal2124 <int>, afatal <dbl>, pop <dbl>, pop1517 <dbl>,
## #   pop1820 <dbl>, pop2124 <dbl>, milestot <dbl>, unempus <dbl>,
## #   emppopus <dbl>, gsp <dbl>, rate <dbl>
```

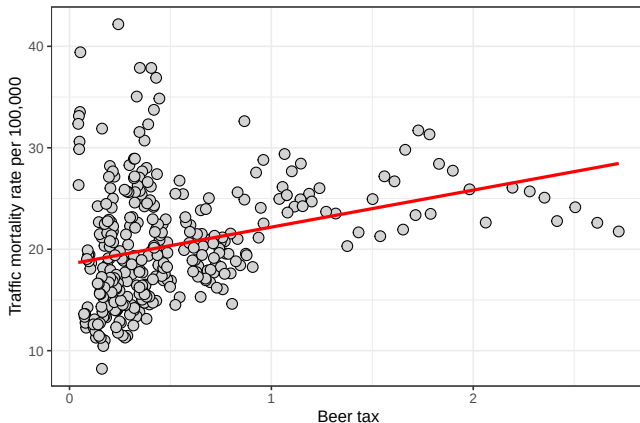
PANEL DATA

$$\text{Rate}_{i,t} = \beta_0 + \boxed{\beta_1} \text{Tax}_{i,t}$$



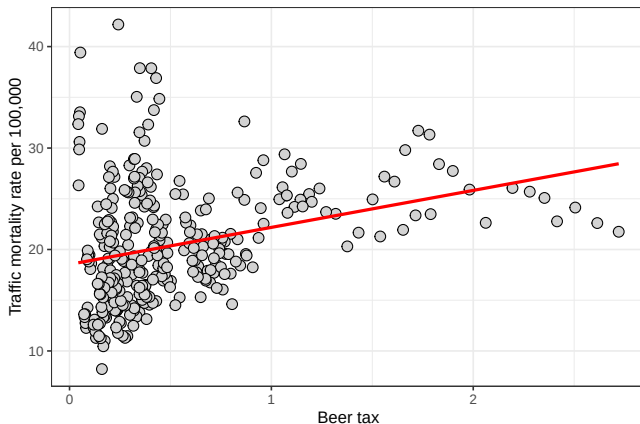
PANEL DATA

- What variation are we using to estimate the coefficient?



PANEL DATA

► Between-unit/time variation.



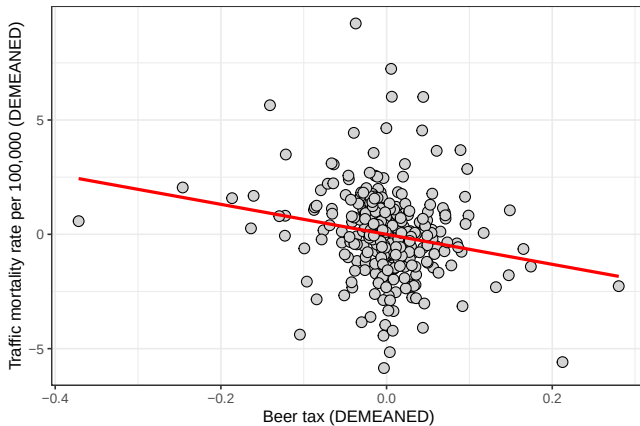
What variation are we using once we control for state?

$$\text{Rate}_{i,t} = \beta_0 + \boxed{\beta_1} \text{Tax}_{i,t} + \beta_2 \text{State}_i$$

PANEL DATA

```
Fatalities <- Fatalities %>%  
  group_by(state) %>%  
  mutate(rate_demeaned = rate - mean(rate),  
         tax_demeaned = beertax - mean(beertax))  
mod <- lm(rate_demeaned ~ tax_demeaned, data=Fatalities)
```

PANEL DATA



- ▶ What variation are we using to estimate this coefficient?

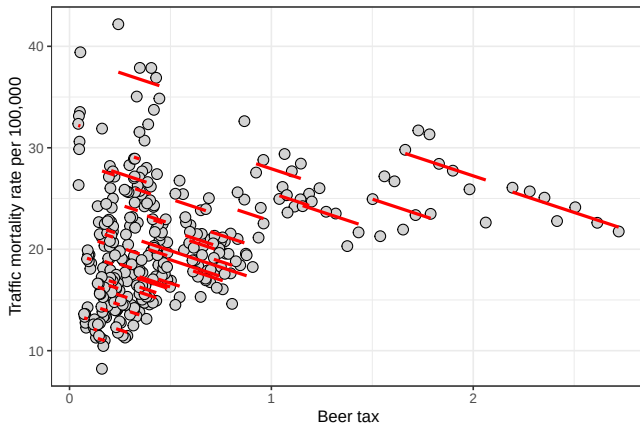
PANEL DATA

- ▶ What variation are we using to estimate this coefficient?
- ▶ Because we controlled for state, we are now using **within-unit** variation over *time* to estimate slope.



PANEL DATA

- Intercept shift by *unit* (state).



PANEL DATA

```
mod <- lm(rate ~ beertax + state, data=Fatalities)
```



PANEL DATA

```
##
## Call:
## lm(formula = rate ~ beertax + state, data = Fatalities)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.8696 -0.8284 -0.0127  0.7955  8.9780
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   34.7763     3.1336   11.098 < 2e-16 ***
## beertax       -6.5587     1.8785   -3.491 0.000556 ***
## stateaz       -5.6773     2.6667   -2.129 0.034107 *
## statear       -6.5495     2.1902   -2.990 0.003028 **
## stateca      -15.0947     3.0435   -4.960 1.21e-06 ***
## stateco      -14.8428     2.8735   -5.165 4.50e-07 ***
## statect      -18.6226     2.8053   -6.638 1.58e-10 ***
## statede      -13.0760     2.9395   -4.448 1.24e-05 ***
## statefl      -2.6813     1.3933   -1.924 0.055284 .
## statega       5.2460     1.8395    2.852 0.004661 **
## stateid      -6.6902     2.5797   -2.593 0.009989 **
## stateil      -19.6162     2.9150   -6.730 9.23e-11 ***
## statein      -14.6154     2.7254   -5.363 1.69e-07 ***
## stateia      -15.4393     2.5344   -6.092 3.58e-09 ***
## stateks      -12.2322     2.4544   -4.984 1.08e-06 ***
## stateky      -12.1752     2.8717   -4.240 3.02e-05 ***
## statela      -8.4712     1.8867   -4.490 1.03e-05 ***
## stateme     -11.0795     1.9112   -5.797 1.78e-08 ***
## statemd     -17.0644     2.8322   -6.025 5.17e-09 ***
## statemg     -21.0975     2.7610   -7.641 3.24e-13 ***
## stateml     -14.8453     2.3602   -6.290 1.18e-09 ***
## statemv     -14.8453     2.6509   -7.157 6.92e-12 ***
```

PANEL DATA

```
Fatalities <- Fatalities %>%  
  group_by(state) %>%  
  mutate(rate_demeaned = rate - mean(rate),  
         tax_demeaned = beertax - mean(beertax))  
mod <- lm(rate_demeaned ~ tax_demeaned, data=Fatalities)
```

PANEL DATA

```
##  
## Call:  
## lm(formula = rate_demeaned ~ tax_demeaned, data = Fatalities)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -5.8696 -0.8284 -0.0127  0.7955  8.9780   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  3.348e-16  9.601e-02   0.000 1.000000      
## tax_demeaned -6.559e+00  1.741e+00  -3.767 0.000196 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 1.76 on 334 degrees of freedom  
## Multiple R-squared:  0.04074,    Adjusted R-squared:  0.03787   
## F-statistic: 14.19 on 1 and 334 DF,  p-value: 0.0001956
```

INTERCEPT SHIFTS

Penguin-level analysis of bill length and depth (β_1):

$$\text{depth}_i = \beta_0 + \beta_1 \text{length}_i + \beta_2 \text{species}_i$$

State-level analysis of traffic deaths and taxes (β_1):

$$\text{rate}_{i,t} = \beta_0 + \beta_1 \text{tax}_{i,t} + \beta_2 \text{state}_i$$

- ▶ With panel data, we're basically doing the same sort of model as before - but now we are controlling for the *unit* of analysis itself.

INTERCEPT SHIFTS

Penguin-level analysis of bill length and depth (β_1):

$$\text{depth}_i = \beta_0 + \beta_1 \text{length}_i + \beta_2 \text{species}_i$$

State-level analysis of traffic deaths and taxes (β_1):

$$\text{rate}_{i,t} = \beta_0 + \beta_1 \text{tax}_{i,t} + \beta_2 \text{state}_i$$

- ▶ With panel data, we're basically doing the same sort of model as before - but now we are controlling for the *unit* of analysis itself.
- ▶ **Next week:** implications for *estimation* (i.e., many units) and *uncertainty* (i.e., correlated errors).

CAUSAL INFERENCE

- ▶ What might panel data and within-unit variation help us do from a *causal* perspective?



CAUSAL INFERENCE

- ▶ What might panel data and within-unit variation help us do from a *causal* perspective?
- ▶ Let's think about trying to estimate a causal effect using **between-unit** variation, as we discussed last week.



AVERAGE TREATMENT EFFECTS

$$y_i = \beta_1 X + \epsilon_i$$

$$\beta_1 = \sum_n (y_i | X = 1) - (y_i | X = 0)$$

- But when can these **predictions** be interpreted as **potential outcomes**?

AVERAGE TREATMENT EFFECTS

$$y_i = \beta_1 X + \epsilon_i$$

$$\beta_1 = \sum_n (y_i | X = 1) - (y_i | X = 0)$$

- ▶ But when can these **predictions** be interpreted as **potential outcomes**?
- ▶ We can't predict individual treatment effects, but we can predict the **average treatment effect (ATE)**. The regression model is one way of generating the **predictions** needed to estimate the ATE.

AVERAGE TREATMENT EFFECTS

Independence assumption

$$(Y^1, Y^0, \delta) \perp\!\!\!\perp X$$

- ▶ Also referred to as assuming exchangeability, no unobserved confounding, etc.

AVERAGE TREATMENT EFFECTS

Independence assumption

$$(Y^1, Y^0, \delta) \perp\!\!\!\perp X$$

- ▶ Also referred to as assuming exchangeability, no unobserved confounding, etc.
- ▶ What do we assume is exchangeable?

CAUSAL INFERENCE



Prince Charles

- Male
- Born in 1948
- Raised in the UK
- Married twice
- Lives in a castle
- Wealthy & famous



Ozzy Osbourne

- Male
- Born in 1948
- Raised in the UK
- Married twice
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- Wealthy & famous

CAUSAL INFERENCE

- Implication: there is always unobserved confounding, which might be really consequential when estimating effects using **between-unit** variation from observational data.



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CAUSAL INFERENCE

- But what if we had longitudinal panel data? What if we can estimate effects using **within-unit** variation?



Prince Charles

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Ozzy Osbourne

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CAUSAL INFERENCE

► Panel data and causal inference

$$\text{Rate}_{i,1988} = \beta_0 + \beta_1 \text{Tax}_{i,1988} + \beta_2 Z_i$$

$$\text{Rate}_{i,1982} = \beta_0 + \beta_1 \text{Tax}_{i,1982} + \beta_2 Z_i$$

$$\text{Rate}_{i,1988} - \text{Rate}_{i,1982} = \beta_0 + \beta_1 (\text{Tax}_{i,1988} - \text{Tax}_{i,1982})$$

CAUSAL INFERENCE

- ▶ Panel data and causal inference: remember the relationship to demeaning! We are removing the **unit-specific means**; in other words, all **time-invariant** characteristics of units.

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- ▶ Why is this magical? Why is this different from the penguin example where we had to *observe* and control for species?
- ▶ Because we are removing lots of characteristics that are **unobserved**!

CAUSAL INFERENCE



Prince Charles

- Male
- Born in 1948
- Raised in the UK
- Married twice
- Lives in a castle
- Wealthy & famous



Ozzy Osbourne

- Male
- Born in 1948
- Raised in the UK
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 - ▶ For example, if $Y \rightarrow X$, then unit-level fixed effects will over-control the mediating pathway by which part of the total effect of $Y \rightarrow X$ over the period operates through $X_{i,1} \rightarrow Y_{i,2} \rightarrow X_{i,3} \rightarrow Y_{i,4}$

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 - ▶ What predicts differences in the average maternal mortality rate across hospitals? Even if I have panel data on hospitals over time, I may not want to include a hospital-level fixed effect. I am more interested in comparing hospitals than I am in comparing changes over time within hospitals.

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Caveats:

- ▶ (1) Unit-level fixed effects cannot address **time-variant** unobserved confounding.
- ▶ (2) Unit-level fixed effects can lead to substantial bias if there is strong reverse causality.
- ▶ (3) Including fixed effects isn't just a "better" way to estimate the same comparison; it *changes* the comparison.

If you are interested in further developing your causal inference skills, I teach a course called **SOC8890: Applied Causal Inference**.